

Assignment 1: Intro

In popular articles such as [Weather.com](https://www.weather.com)'s "Too hot for takeoff," the primary narrative for concern is the possibility of seeing delayed or cancelled flights. The problem, as explained, is that as global warming starts to increase the mean surface temperature, the air becomes thinner and airlines must subsequently delay the flight until cooler, denser air arrives at the airport. The fact sheet provided by ICAO lists several reasons why this occurs: not only is the air thinner and the wings generate less lift, but the engines also produce lower amounts of thrust to weight, a major component in takeoff. This compounding effect makes it even more difficult for planes to takeoff. Many airports with shorter runways must reduce weight in order to takeoff in summertime (YCC Team).

Both journal articles from Coffel and Horton use the reduction in weight limits as their primary outcome measure, measuring how many days airports with specific models will need to reduce weight by 5 or 7.5 tons, corresponding to roughly 52 or 79 passengers with luggage. This outcome by weight reduction days is plotted for 4 specific models of airplanes matched against 19 at-risk airports. At risk includes short runways such as New York's Lagoon, and high-altitude runways such as Denver International.

Both articles go into detail about a loss of airline revenue due to the weight restriction, but neither mention how it might affect what passengers could care about: cost. The first study pertained to a Boeing 737-800. Accounting for a 7.5 ton weight reduction leads to a 30% drop of its actual payload capacity. Ticket prices for at-risk airports could jump an additional 42.8% (linear, not including lessened fuel requirement).

Another gap is the potential for airlines to start over-scheduling at-risk airports during morning and evening hours. Popular evening runway times can become crowded and necessitate moving flights to unpopular red-eye times. Another possible measurable outcome here is how consumer sentiment may negatively view these unpopular flight times. One way to potentially measure this is to define a *comfortable* consumer time for flights (not too early, nor too late), and calculate the number of airplanes that would be unable to takeoff or land during those hours without any weight reductions. The resulting plot could be a colored world map detailing the number or percentage of that airport's flights that will need to be delayed or cancelled.

Airplane model data is available from manufacturers already ("Characteristics for Airport and Maintenance Planning"), but the plots come in weight-vs-runway-length form, with various contour lines detailing altitude. Boeing inverts their plots from Airbus, but having runway length as the y-axis makes the most sense. The plots come with a standard called *ISA* and *ISA+15* form, which have some temperature normalization to it. I will first find a curve fit for all contour lines and unpack the ISA formula to eventually get a runway-length-vs.-air-density type plot for each airframe. Once that can be established, writing a program to use this data in an automated way should not be difficult. Getting the data into a system and curve fitting will be a trickier process than initially thought.

References

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